

Assignment 1: Maze Solver using A* Algorithm with Movement Constraints

A Python script with an AI algorithm that solves a 2D maze using the A* search algorithm - but, with specific movement constraints, to really force the AI earn its way out of the maze. This time, after first entering the maze from the bottom right corner, the AI can only move forward, or turn right and then move forward. It can turn left but cannot move backward.

The idea was to show its problem-solving ability to navigate complex environments even with limited movement options.

This solver print-outs its steps into the console during runtime, and in the end also creates a graphic image of the maze with the successful path eventually found.

PRE-INSTALLATION:

pip install matplotlib numpy (if you use python)

MOVEMENT CONSTRAINTS

📌 Allowed Actions:

- o Move Forward: The agent moves one cell forward in the direction it is currently facing.
- o Turn Right, left and Move Forward: The agent turns 90 degrees to the right and left and then moves one cell forward in the new direction.

📌 Disallowed Actions:

- o Move Backward
- o Turn in Place Without Moving

A* SEARCH ALGORITHM

The A* algorithm is used to find the optimal path from the start to the end point within

the maze, considering the movement constraints.

- 📌 Nodes: Each node represents a state defined by its position and the direction the agent is facing.
- 📌 Open List: Nodes to be evaluated.
- 📌 Closed List: Nodes already evaluated.
- 📌 Cost Functions:
 - o $g(n)$: The cost to move from the start node to node n .
 - o $h(n)$: The heuristic estimated cost from node n to the goal (Euclidean distance).
 - o $f(n)$: Total cost of the node ($f(n) = g(n) + h(n)$).

HEURISTIC FUNCTION

- 📌 Euclidean Distance: Used to estimate the cost from the current node to the goal:

MAZE REPRESENTATION

- 📌 The maze is a 2D list where:
 - o 0 represents a walkable path.
 - o 1 represents a wall.

VISUALIZATION

- 📌 Matplotlib is used to create a visual representation of the maze and the agent's path
- 📌 The path is plotted over the maze image, with the start point marked in green and the end point in blue:
 - o Walls are shown in black.
 - o Walkable paths are shown in white.
 - o Agent's path is overlaid in red.
 - o Start point is marked in green.
 - o End point is marked in blue.